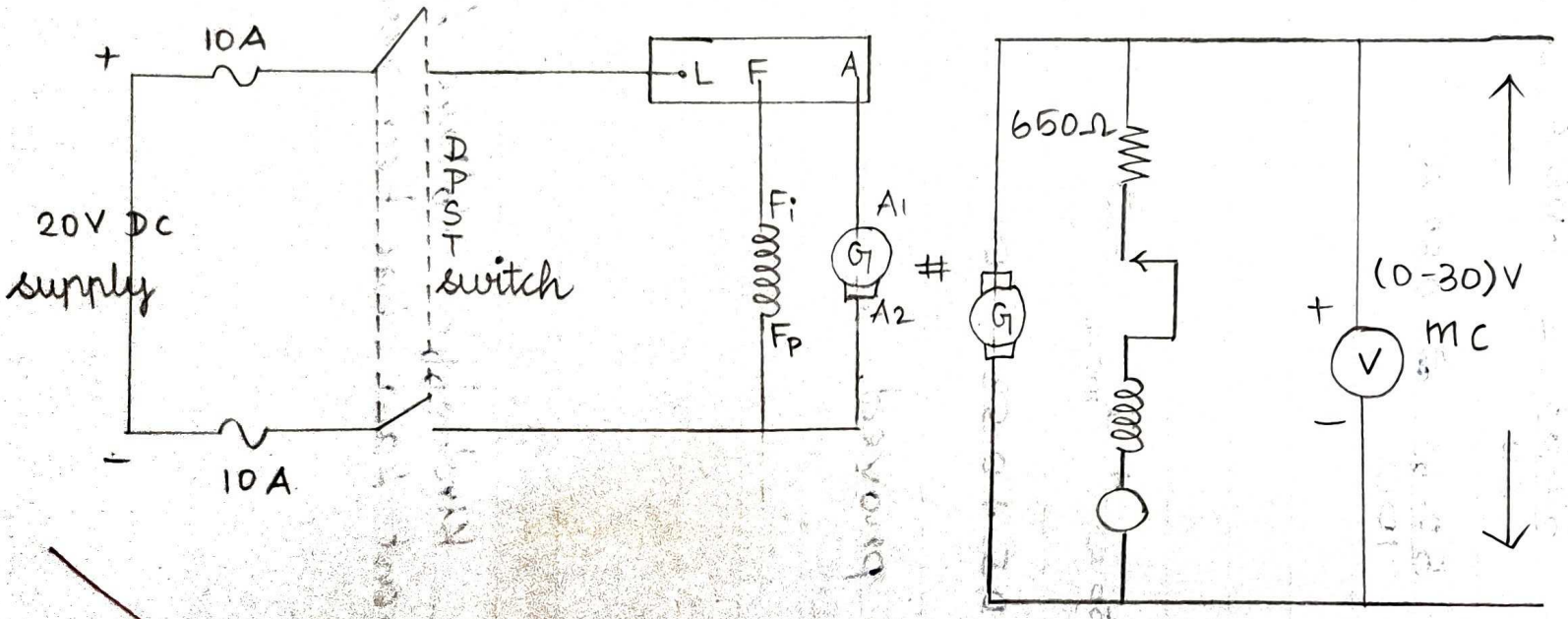
OPEN CIRCUIT CHARACTERISTICS



OPEN CIRCUIT AND LOAD CHARACTERISTICS OF DC SHUNT GENERATOR.

AIM:

To draw the open circuit and load characteristics of DC shunt Generator.

APPARATUS REQUIRED:

Laptop with internet connection

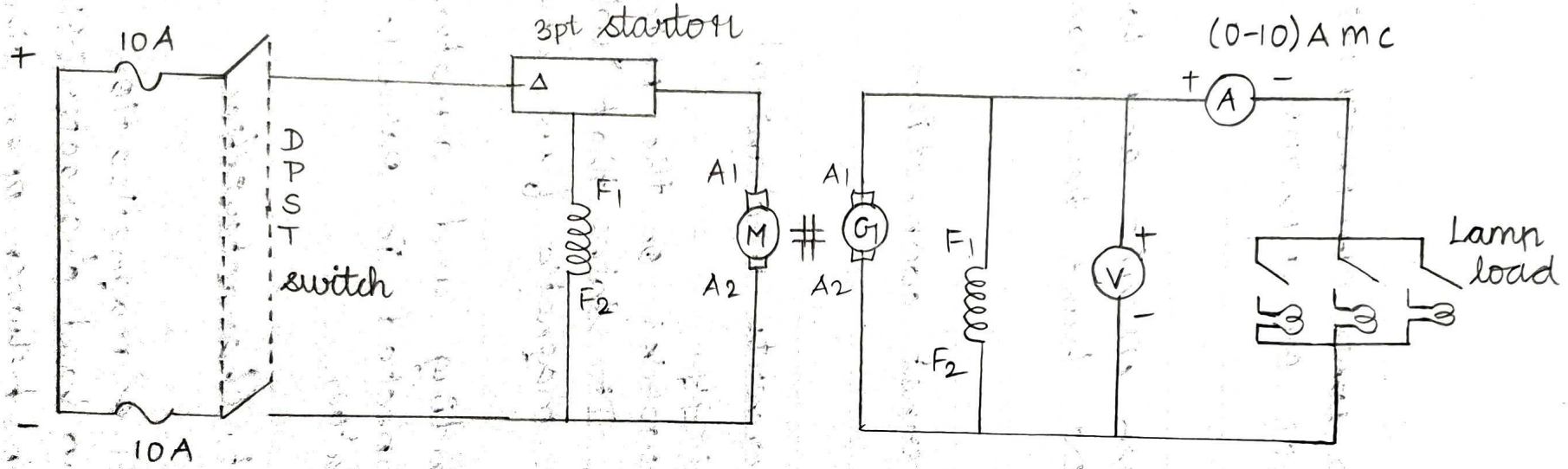
THEORY:

* DC generator converts mechanical energy into electrical energy.
* DC generator works on the principle of faraday's law. It states that "when a conductor cuts the magnetic flux and emf is induced."
* The field winding and armature windings are connected in parallel in the shunt generator.

Open Circuit Characteristics

It is also known as magnetization characteristics or no load characteristics. It is a graph drawn between open circuit voltage (E_0) and field current (I_f).

LOAD
~~OPEN~~ CIRCUIT CHARACTERISTICS



Load circuit Characteristics

It is a graph drawn between load voltage (V_L) and load circuit (I_L). It is also known as external characteristics.

PROCEDURE:

Open Circuit Characteristics:

1. Connections are made as per the circuit diagram.
2. Switch on the supply and note down the readings (voltage and current)
3. By varying the field rheostat different values of open circuit voltage and field current are noted.
4. Plot the graph between open circuit voltage and field current.

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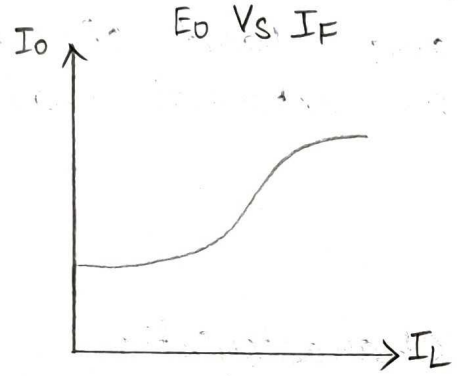
OPEN CIRCUIT CHARACTERISTICS

MODEL GRAPH :



S.No	Voltage(V)	Current (A)
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.21
5	133	0.23
6	135	0.24
7	138	0.28
8	142	0.30

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Load Circuit Characteristics:

1. Connections are made as per the circuit diagram.
2. Switch on the supply and note the no load voltage and current.
3. By connecting different nodes, different values of load voltage and current.
4. Plot the graph between load voltage and load current.

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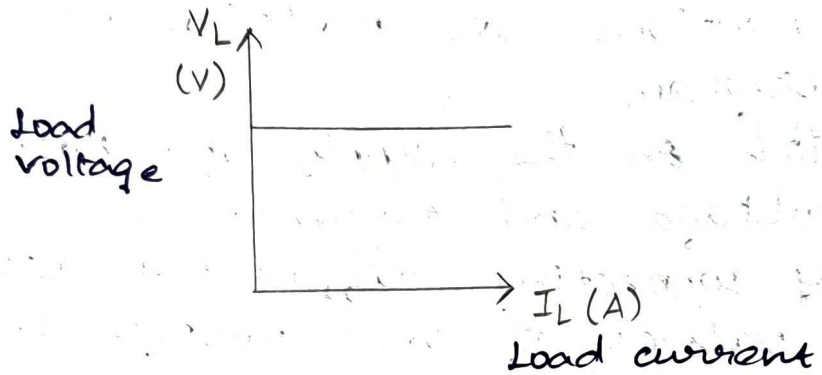
LOAD CIRCUIT CHARACTERISTICS

MODEL GRAPH



S.No	Current(A)	Voltage (V)
1	1.2	220
2	2.8	212
3	3.2	208
4	3.6	205
5	5.5	200
6	7	195

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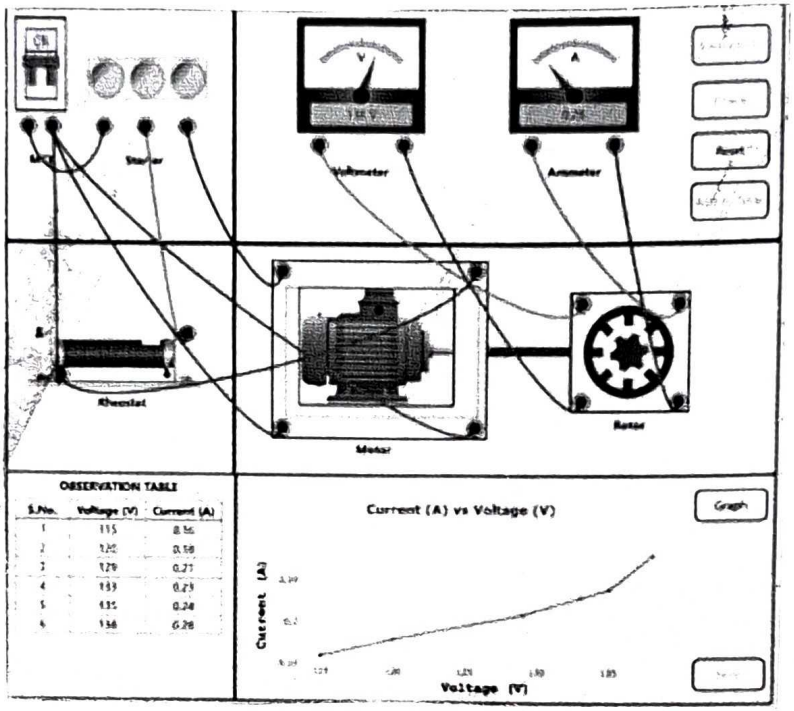
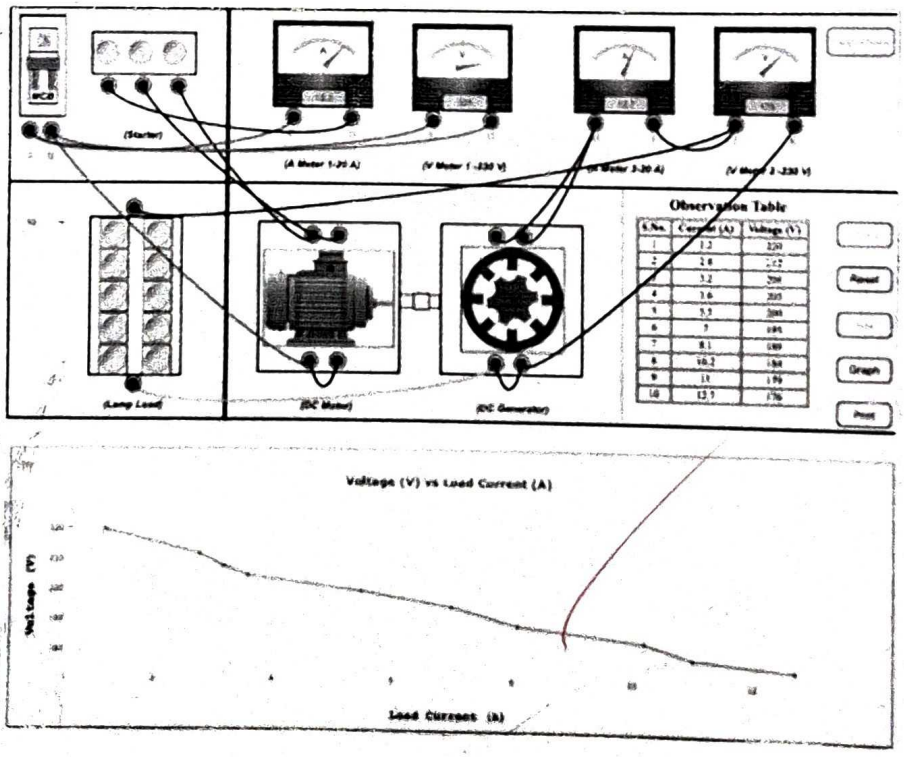
RESULT:

Thus the open circuit and load characteristics of DC shunt generator were verified successfully.

OPEN CIRCUIT CHARACTERISTICS



LOAD CIRCUIT CHARACTERISTICS: